

A Modularized Sensitive Skin for Motion Planning in Uncertain Environments

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Abstract

One way to avoid obstacles without modeling the environment is to use a sensor-based sensitive skin covering the body of the robot along with proper sensor-based motion planning algorithms. Compared to prior attempts, the system developed at the UW-Madison Robotics Lab and discussed here presents a system that can be easily reconfigured for a variety of applications. This is accomplished via modularized sensitive skin patches that can be connected in various ways, and a hierarchical control architecture that can be easily modified for various system configurations. In the implemented version, the skin contains about 1000 infrared sensors; it can be easily expanded, with minor changes. The system has been successfully tested in preliminary trials.

1 Introduction

One of the goals in robotics is to endow robots with the ability to move and operate autonomously in an uncertain environment. Using this ability, the robot would be able, for example, to operate in an environment with unknown, perhaps moving, obstacles. The primary methodology to achieve this goal is sensor-based motion planning. One well-known option for the hardware front-end of such a system is a sensitive skin that covers, for example, the whole body of an arm manipulator. With this skin (based e.g. on tactile or proximity sensing) the robot can detect surrounding objects and, using appropriate sensor-based motion planning algorithms, move purposely in its workspace.

There have been a number of types of sensor systems reported in literature for this purpose, each with own advantages and drawbacks. None except [1] provide an ability to cover large surfaces of the robot and thus endow it with a sensory ability similar to that of humans and animals (which is indispensable in a truly uncertain environment). Capacitive-coupling proximity sensors [2] are reliable and robust; their drawback

is inferior resolution - i.e., since the sensor cannot resolve the object dimensions within its (fairly large) sensing radius, the robot has to assume the object to be large. E.g., to be on the safe side, a 2cm object will be perceived as 20-30cm object; this may dramatically limit the robot arm's room for maneuvering. Such sensors are also bulky and, at least today, not feasible for large continuous patches of skin; their detection properties vary widely when dealing with conduction vs dielectric materials. Similar pluses and minuses can be attributed to the sensors based on magneto-inductive effects [3], and to ultra-sound sensors. Laser range finders are excellent as obstacle detection devices, especially for mobile robot navigation [4] [5]. Potentially miniaturization may bring them to the level when arrays of such devices may form continuous skin patches.

With all the conflicting demands imposed on the sensitive skin design, at this point infrared (IR) sensors seem to present the best compromise for skin designs. They are fast, relatively small, and have good resolution. The surface-mounted versions provide further miniaturization possibilities. Similar to vision, IR performance does not depend on the object's material. Their disadvantages lie mainly in the inability to provide distance measurements at short distances, and some sensitivity to object's color and texture.

While the first generation IR-based skin demonstrated validity of the underlying motion planning principles [1], the customized design of the skin it offered prevented it from demonstrating an important premise - that it can be a truly universal means of motion control. That is the topic of the current work.

Clearly, from the standpoint of motion planning, the sensitive skin principle should be applicable to any moving machine, be it an arm manipulator, a mobile robot, or a driverless/teleoperated automobile [6]. While in each of these cases the control system needs to utilize specific knowledge of the machine's kinematics, its sensitive skin system can be roughly the same, provided its material and software architecture are flexible enough to fit various body and control configurations.

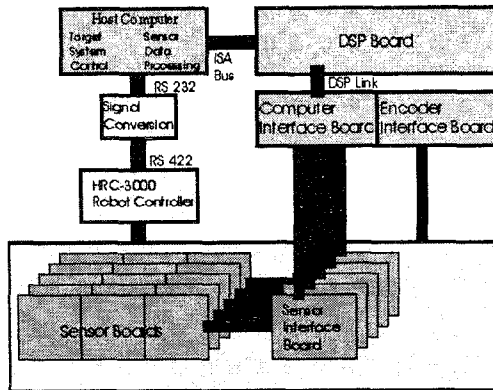


Figure 1: Hardware block diagram.

This requirement of flexibility calls for modular, easily connectable skin patches - such that they could be cut to appropriate sizes and shapes without losing their control properties. It also calls for hierarchical control architecture capable of handling various numbers of skin boards. A high-level diagram of our modular sensitive skin system is shown in Fig. 1. Each Sensor Board includes an array of sensors, each of which is a surface-mounted IR LED/detector pair.

Initially all boards are identical 20cmx20cm square boards; to fit in place, some of them may be cut by scissors to the shape that satisfies the customer's size and shape requirements. Boards are made of Capton material flexible enough to comply with the shape of the robot body. They do not stretch, thus not endangering the board wiring. Each sensor board can be thought of as a section of skin - just as human skin has sensors (nerve endings) distributed across its surface, so too does the robot skin.

The sensor density on the skin (i.e. the spacing between the sensors - here 2.5cm) determines the resolution of objects that the skin detects. That is, if only one sensor detects an object, the motion control system will have to assume, to be on the safe side, that the object is at least 2.5cm long. While this loss of resolution is not very important for a mobile robot, it may be quite important for an arm manipulator, due to the nonlinear relations between the object size and the arm links rotation necessary to pass around it. Sensor Boards have various connectors, allowing one to connect them together on the robot's body to form a continuous surface, or regroup them into a new configuration, to accommodate design changes.

The lower-level control of the skin Sensor Boards is via Sensor Interface Boards, whose purpose is to address sensors and decode sensor information. This information is then sent to a processor that houses

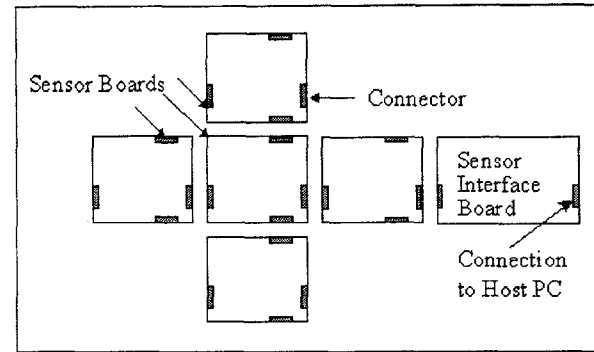


Figure 2: Connections between the Sensor Boards and the Sensor Interface Board.

control and display software. Similar to the Sensor Boards, Sensor Interface Boards are modularized; each serves a few Sensor Boards, Fig. 2. Such architecture makes the lower-level control independent of the number of sensors in the skin, which allows one to reconfigure easily the control software and helps improve the sensing speed and performance. The third element in the control chain, the Computer Interface Board, provides the interface between the Sensor Interface Boards on the skin and the host computer processors. The next two sections present in more detail the sensitive skin system designed and built at the UW-Madison Robotics Lab. The said three components of the skin are discussed in Section 2, followed by the implementation issues in Section 3. For details on motion planning algorithms, refer to [6].

2 Skin Control Components

Sensor Board. As discussed above, the sensitive skin is modularized by breaking it into smaller pieces of standard configuration. Various geometric shapes were considered for the standard sensor board, to allow for more flexibility and simplicity in the design, assembly, and maintenance stages. The shape chosen is a 20cmx20cm square. In general, rectangular shape is preferable for engineering purposes; a square's diagonal and side symmetry offers additional possibilities for sensor board combinations and customized modification. A flexible Capton material was chosen for the board, to house the sensor pairs and corresponding control logic.

Each sensor board contains 8 columns and 8 rows of transmitter/receiver sensor pairs, for the total of 64 pairs. Sensors are wired in parallel such that all 8 sensors in a row are polled at the same time. This re-

duces the number of circuit connections and the data acquisition time. The disadvantage of this scheme is that since 8 sensors are active at the same time, there is a chance that the transmitter signal from one sensor pair is reflected by an object and received by the receiver of another sensor pair. In practice this ambiguity is rare, as most objects' matte surfaces reflect most of signal energy to the (closest) receiver of its own sensor pair.

If due to a particular design a board must house fewer than 8x8 sensors, the standard row/column structure is still kept intact, and the number of sensors per row is decreased accordingly. This allows for shapes like an approximation of a circle or a trapezoid. To provide connections between the sensor and the Sensor Interface boards the former have connectors on all four sides as shown in Fig. 2. A group of Sensor Boards plus one Sensor Interface Board that controls them form a Sensor Module. The number of Sensor Boards that can be grouped in one Sensor Module is limited by the driving current of the sensors.

To control the power dissipation of a Sensor Board, a special row/column architecture minimizes power consumption. One LED consumes about 1.2 Watts, with the total of about 10 Watts per LED column. That is, at a given moment in a given Sensor Module only one sensor column is fired at once, giving a power drain of 10 Watts for IR light firing per Sensor Module. Consider, for example, a system with five full Sensor Modules (which is sufficient to cover a typical industrial robot arm); the total number of sensors is then $5 \times 3 \times 64 = 960$. The total power for column control for the entire skin is thus 50 Watts. The IR receivers are similarly connected in parallel by rows. Each row consumes 0.6 Watts, making it about 5 Watts per Sensor Module, or 25 Watts for the system. This brings the total instantaneous power needed to $50 + 25 = 75$ Watts. A drawback to the described addressing scheme is that access to specific sensor pairs can only be achieved in a sequential fashion.

Sensor Interface Boards. These boards control low-level processing of the digitized sensor data from the corresponding Sensor Boards. Each Sensor Interface Board controls a set of Sensor Boards; together, they form a Sensor Module.

A system with five Sensor Modules thus has five Sensor Interface Boards; with three Sensor Boards per module, this amounts to 15 Sensor Boards in total. As mentioned before, such a setup is fully reconfigurable depending on an application. Sensor information from Sensor Modules is processed in parallel, resulting in an efficient data transfer. Physically, a Sensor Interface

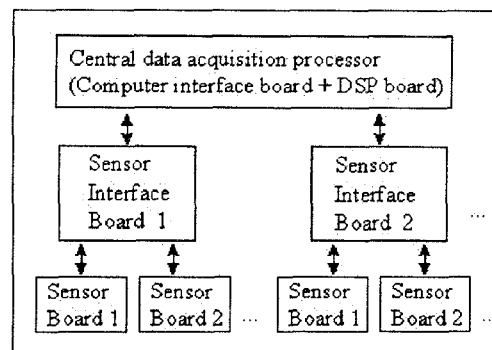


Figure 3: The hierarchical Skin architecture.

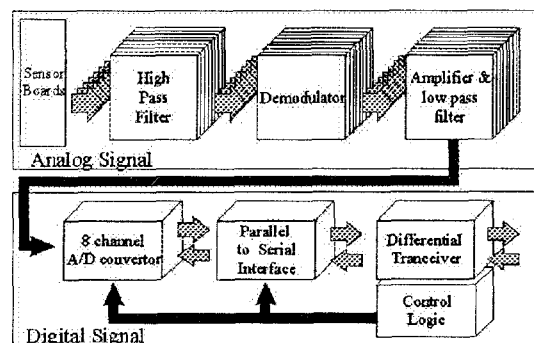


Figure 4: Sensor Interface Board diagram.

Board is built on a less flexible and smaller (compared to Sensor Boards) Capton board; it is placed underneath the skin, closer to the Sensor Boards it serves. The power lines and signal lines are passed underneath the Sensor Boards. The hierarchical structure of the skin hardware architecture is shown in Fig. 3.

A diagram of a Sensor Interface Board is shown in Fig. 4. To improve the performance, ambient light is filtered out using a high-pass filter. Then the signal is demodulated using different frequencies for the sensors on different links, so that each sensor reacts only to the light from LEDs on its own link. Finally, further noise is reduced and the signal is amplified before the A/D conversion.

For the entire array of sensors (in our case, 5 Sensor Modules, about 1000 sensors in total) our goal has been to achieve a 20ms sampling cycle required by a typical industrial arm manipulator control system. This is done via transputer technology: a pair of transputer link chips capable of transmitting data at a rate of about 10 Mbits per second deliver the desired performance [7]. The communication channel between the Computer Interface Board and Sensor Interface Boards is composed of 5 independent and par-

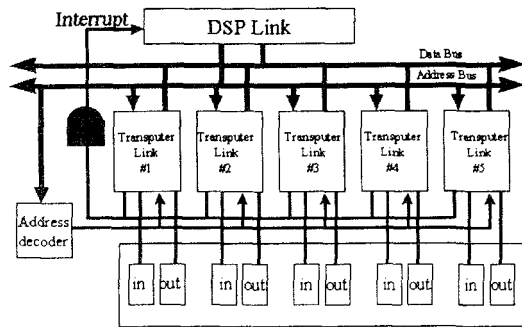


Figure 5: Computer Interface Board diagram.

allel Link Adapters, each of which is responsible for one Sensor Module. The output bytes of the communication link chips are used to command the A/D converter and the shift register located on each Sensor Board.

Computer Interface Board. As a part of the central data acquisition processor, the Computer Interface Board plays an important role as a gateway between the Sensor Interface Boards and the DSP Board. In order to process the skin sensor data, there are as many communication link processors as the number of modules. Each link processor can talk to each module independently, thus allowing parallel data processing. The Computer Interface Board interacts with the DSP Board through the DSP link, achieving a high data throughput, with the rate up to 5 Mword/sec. It also contains the interface between the robot arm encoders and the DSP Board. A schematic diagram of the Computer Interface Board is given in Fig. 5. There are five serial connection links for the transputer link pairs, each with 8 bit parallel-to-serial conversion capability. This scheme allows massively parallel data transmission between two remotely located system units, while maintaining noise immunity.

3 Implementation Issues

Physical Design. In this work, the modularized sensitive skin has been implemented on a Hitachi M6100 industrial robot arm, Fig. 6. The arm has six degrees of freedom and is powered by AC servo motors. Linkage synchronization is provided by absolute address encoders. The access to the arm control system is accomplished by simulating the arm teaching pendant (a tool used for manual robot arm programming and testing) in the host PC software. In summary, the system includes these components:

- 1) Encoders: read the arm joint angles.

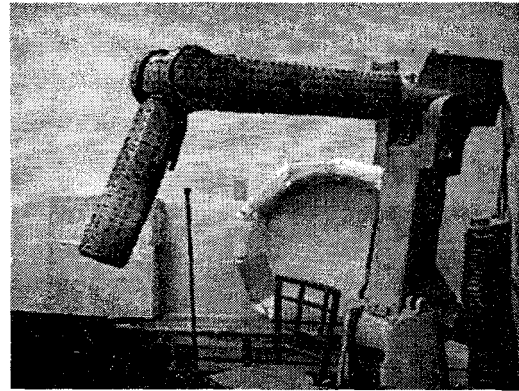


Figure 6: Modularized Sensitive Skin installed on an industrial Robot Arm.

- 2) Sensor Boards.
- 3) Sensor Interface Boards.
- 4) Computer Interface Board: interface between sensor data processing units and arm controller.
- 5) DSP Board - contains processors that perform motion planning and other signal processing tasks.
- 6) Host Computer: houses various boards, provides the operator interface/development environment.
- 7) Signal Conversion Board: handles the arm controller-host PC communication.

The host PC issues motion commands based on the sensor data processed by the DSP Board and the motion planning algorithms (see [6] for details). In terms of motion planning, the software includes several layers, which can be roughly divided into two parts: sensor data processing, and motion planning. Due to the skin modularization, the sensor data processing mechanism residing on the DSP Board does not need to be changed when modifying the skin configuration. A low-level data processing algorithm occupies one of the DSP processors; local and global motion planning algorithms reside on the other processors. They are programmed based on the parallel computing environment provided by the DSP programming library [9].

In the current version, the skin covers two out of the three major links of the arm, Fig. 6. When designing the skin mechanical installation, these issues were considered: (a) simplicity of attachment and detachment, (b) flexibility for maintenance, (c) physical tightness for dynamic motion. There are three layers in the mechanical attachment structure (Fig. 7).

The first layer, which is in physical contact with the arm, is a rubber pad that wraps around the robot body. This layer helps accommodate possible bumps, like bolts or rivets, on the surface of the arm, so that the Sensor Interface Boards can be positioned more

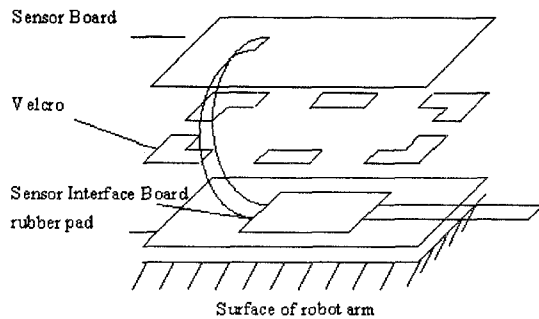


Figure 7: Mechanical architecture of the Sensor Boards.

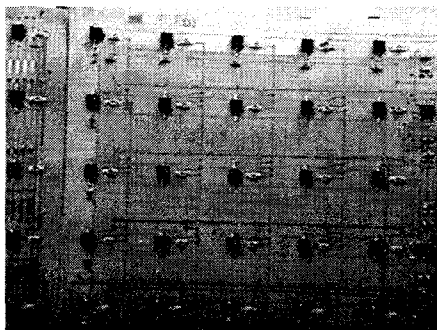


Figure 8: Close-up photo of the Sensitive Skin.

easily on the arm. Sensor Boards occupy the next level, which also contain Velcro strips that form cavities for the Sensor Interface Boards and for electric wires that go underneath the Sensor Boards. Velcro strips are strong enough to hold the Sensor Boards in place, and convenient enough to assure quick attachment and detachment necessary during troubleshooting or skin reconfiguration.

Close-up photos of skin sections are shown in Fig. 8 and Fig. 9. Note the surface-mounted sensor transmitters and receivers, positioned in a grid fashion on the flexible Sensor Boards. Note also the round Sensor Board in Fig. 9, which was cut to accommodate the shape of the arm link. This reshaping of the board without losing its functionality is made possible by the modular design of the board.

Data Acquisition. The software, all developed in the UW-Madison Robotics Lab, is layered: the layers can be roughly grouped into the Application Layer and the Library Layer, as shown in Fig. 10. The Application Layer handles the user interface and data display, and also parameters specific to the current skin implementation, such as (a) the number of sensor modules, (b) the number of Sensor Boards per module, (c) the number of sensors per Sensor Board, (d) the shape of

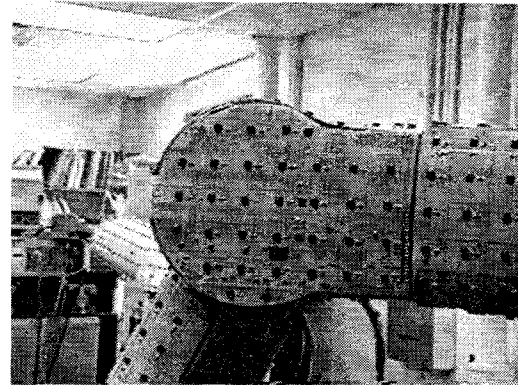


Figure 9: Sensor Board fitted to a round section of the Arm.

each Sensor Board, (e) the sensor pair arrangement.

The Library Layer contains components for motion planning and skin configuration-independent parameters (such as the relation between the number of sensors specified in the Application Layer and the data units from Sensor Interface Boards). The robot control library allows testing of various motion patterns with the arm links/joints, together or separately. The implemented modularity in software architecture reduces the time needed to convert one skin configuration to another - only modules in the Application Layer need to be modified to reflect the new configuration.

Modularity in the skin design also allows the use of parallelism in gathering data from the skin. Data processing is done in only one CPU processor, via CPU time-sharing for five independent Sensor Interface Boards. The implemented time-sharing mechanism is based on Round Robin scheduling [8], one of several existing non-prioritized multi-tasking scheduling algorithms. Fig. 11 shows the parallel task scheduling scheme. For example, a data processing command is sent through the data bus to the Sensor Modules 1 and 2 exactly at the same moment. During the next time slice the scheduler will handle the data processing for Sensor Modules 3 and 4. This scheduling mechanism speeds up data processing significantly - as much as by a factor of 10 - compared to the serial data processing system described in [1].

System Test. In the first trials with the modularized sensitive skin, a Repelling Mode motion planning algorithm has been used to check the overall performance of the current system. As the name implies, when a sensor on the skin detects an obstacle, the arm simply moves away from the obstacle. While this mode is by no means sufficient for sophisticated mo-

Layer	Module	Function
Application	User interface/data display	Command and monitoring data display
Application	Implementation specific skin parameters	System dependent data parameters
Library	Motion planning	Motion planning algorithms
Library	Robot arm control	Software arm control interface
Library	Low level data processing	Skin data processing

Figure 10: Software architecture.

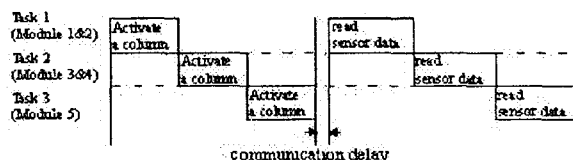


Figure 11: Parallel data processing diagram.

tion planning, it helps test the multiple hardware and software components that constitute the skin architecture. Briefly, the operation of the system during Repelling Mode motions planning proceeds as follows. First, raw sensor data is gathered and processed by one of the four DSP Board CPUs. Meanwhile, the arm joints' encoder signals are stored. The raw sensor data is then filtered and transmitted to the host computer together with the encoder signals, using the corresponding communication protocols. The data is then processed, to make a decision on the direction of motion for the next step, and passed to the arm controller as the output of the Repelling Mode algorithm. The result is a small motion step in three-dimensional space in the direction away from the obstacle, thus producing continuous motion. With the Repelling Mode, the skin system's ability for motion planning has been successfully tested by using hand-held obstacles placed in the path of the robot arm.

4 Summary and Conclusions

A modularized sensitive skin for motion planning has been developed in this project. Tests showed satisfactory performance of the system. This allows us to move to the next stage of the project, in which the described system will be integrated with proper motion planning algorithms. With these algorithms in place, the system is expected to move purposely in a complex environment with still and moving obstacles. The main contributions of the current work are: 1) Modular Approach: the developed design scheme results in a modular skin system which can be easily

reconfigured to meet the requirements of a particular arm manipulator, a mobile robot, or any other machine.

2) Parallel Data Acquisition: a significant polling performance improvement has been achieved by separating the data acquisition and data processing functions. This is done by allocating a separate CPU of the DSP Board for data acquisition, and implementing a parallel data acquisition algorithm which polls the Sensor Boards. As a result, the data processing time has been reduced about 10 times compared to the prior system reported [1].

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